PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

OUTLINES OF TESTS SYLLABI AND COURSES OFREADING

FOR

Bachelor of Vocation (Hardware and Networking)

Session 2017-18

(1st to 6th Semester)

SCHEME OF B.Voc.(HARDWARE AND NETWORKING)

(SEMESTER SYSTEM)

Semester I								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Cr**
*GEN -101	Communication Skills	Generic	Theory	20	80	--	--	
*GEN -102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	
HWN-103	Computer Hardware	Skill	Theory & Practical	10	40	10	40	
HWN-104	Network Fundamentals	Skill	Theory & Practical	10	40	10	40	
HWN-105	Practical on Networking-I	Skill	Practical	-	-	20	80	
SEMESTER II								
*GEN 201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
GC-202	Business Ethics	Generic	Theory	20	80	-	-	6
HWN-203	Personal Computing Software	Skill	Theory & Practical	10	40	10	40	6
HWN-204	Computer Hardware and Trouble Shooting	Skill	Theory & Practical	10	40	10	40	6
HWN-205	Practical on Networking-II	Skill	Practical	-	-	20	80	6
**SIT-201	Summer Industrial Training	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

Job Role: Field Technician-Computing and Peripherals

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within in one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester**

Semester III								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 301	Value Education And Human Rights	Generic	Theory	20	80	--	--	6
GC-302	E-Commerce	Generic	Theory	20	80	-	-	6
HWN-303	Windows Operating System	Skill	Theory & Practical	10	40	10	40	6
HWN-304	Routing and Switching	Skill	Theory & Practical	10	40	10	40	6
HWN-305	Routing Configuration & Network Connectivity	Skill	Practical	-	-	20	80	6
SEMESTER IV								
*GEN 401	Environmental Studies	Generic	Theory	20	80	--	--	6
GC-402	Project Management	Generic	Theory	20	80	-	-	6
HWN-403	Windows Server Administration	Skill	Theory & Practical	10	40	10	40	6
HWN-404	Network Administration & Trouble Shooting	Skill	Theory & Practical	10	40	10	40	6
HWN-405	Switching Configuration & Network Connectivity	Skill	Practical	-	-	20	80	6
**SIT-401	Summer Industrial Training	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted withinin one week of start of 5th Semester.Viva-Voce examination to be held within 3-weeks of the start of 5th semester.**

Job Role: Network Support Engineer

Semester V

Paper Code	Title	Generic / Skill Component	Theory / Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN - 501	Critical Thinking and Elementary Statistics	Generic	Theory	20	80	--	--	6
GC-502	Introduction To Research Methodology AndReport Writing	Generic	Theory	20	80	-	-	6
HWN-503	Linux Operating System	Skill	Theory & Practical	10	40	10	40	6
HWN-504	VMware Workstation	Skill	Theory & Practical	10	40	10	40	6
HWN-505	Advanced Routing and Switching	Skill	Practical	-	-	20	80	6

Note: Winter Industrial/ In-house Training of 2-3 weeks in a relevant area after 5th Semester Examinations in winter break.

SEMESTER VI

*GEN 601	EntrepreneurshipDevelopmentProgramme	Generic	Theory	20	80	--	--	6
GC 602	Total Quality Management	Generic	Theory	20	80	-	-	6
HWN-603	Wireless Communication And Networking	Skill	Theory & Practical	10	40	10	40	18
HWN-604	Firewall Technology	Skill	Theory & Practical	10	40	10	40	6
HWN-605	Implementation Of Advance IP Routing	Skill	Practical	--	--	20	80	6
**SIT-601	Summer Industrial/ In-house Training and Comprehensive Viva	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

****Winter Industrial/ In-house Training of 2-3 weeks done after 5th Semester Examinations and before start of 6th semester.Training report by the student to be submitted within in one week of start of 6th Semester.Viva-Voce examination to be held within 3-weeks of the start of 6th semester.**
Job Role: Network Administrator

B.Voc. (HARDWARE AND NETWORKING) Semester: I

Paper Title: COMMUNICATION SKILLS Paper Code: *GEN 101 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objective: The objective of this paper is to develop communication skills, discover what business communication is all about and learn how to adapt the communication experiences in life and to the business world.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to Communication: Need for Effective Communication, The Process of Communication, Levels of communication, Flow of communication, Use of language in communication, Communication networks, Significance of technical communication, Barriers to Communication, Types of barriers, Miscommunication, Noise, overcoming measures.

UNIT-II

Verbal Communication: Planning, Preparation, Delivery, Feedback and Assessment of

Activities like –Public Speaking, Group discussion, Presentation Skill, Audio-Visual Aids,

Personal interview**. Non-Verbal Communication:** Body Language, Personal Appearance, Posture, Gestures, Facial Expressions, Positive attitude formation, Process of attitude formation, How to build a successful attitude.

UNIT-III

SWOT analysis: Self-management techniques, Self-image and Self-esteem, Building self-confidence, Power of irresistible enthusiasm, Etiquettes, Etiquettes in Social as well as Office Atmosphere, Telephone Etiquettes, E-mail Etiquettes, etiquettes and manners. Importance of listening and responding.

UNIT-IV

Report Writing: Business reports: Types, Characteristics, Importance, Elements of structure, Process of writing, Order of writing, the final draft, check lists for reports. Business.Correspondence: Inviting quotations, Sending quotations, placing orders, Inviting tenders, Sales letters, claim & adjustment letters and social correspondence.

Suggested Resources for Reading:

1. 'Management: A Global and Entrepreneurial Perspective', by Heinz Weihrich, Mark.V.Cannie, Harold Koontz., Tata McGraw Hill Publication
2. 'Developing Communication Skills' by Krishmohan and Meera Banerjee, Macmilan India Ltd.
3. 'Business Communication' by Krizan, Merrier, Jones, Thomson Learning - 6thEdition
- 'Communication Skills' by Sanjay Kumar &PushpLata, Oxford University Press

References Resources:

1. 'An Approach to Communication Skills' by Indrajit Bhattacharya, Delhi : DhanpatRai
2. 'Introduction to Psychology' by Atkinson and Hilgard's, Edward E. Smith, Susan Nolen-Hoeksema, Barbara Fredrickson, Geoffrey Loftus

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Paper Title: FUNDAMENTALS OF INFORMATION TECHNOLOGY Paper Code: *GEN 102Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objective: The objective of the paper is to familiarize the students with developments in Information Technology, Internet and use of computer systems at operating system level and application level.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Computers: Introduction to computers, characteristics of computer, organization of computers, hardware, software, data, information, Types of computer: Classification on the basis of purpose: digital computers, analog computers, hybrid computers; Classification on the basis of size: microcomputers, mini computers, mainframe computers and supercomputers, desktop computers, laptops, workstations, PDA. Generations of computer, Uses and Application of computers.

UNIT-II

Computer Hardware: Input Devices: Keyboard, mouse, light pen, joystick, trackball. Voice input device: microphone; Output Devices Printers: types of printer, Plotters, Speakers; Scanners: types of scanners, Visual display devices. **Computer Storage:** Representation of information, BIT, BYTE, Memory, Types of memory: primary memory RAM: static RAM, Dynamic RAM; ROM, EROM, EPROM, EEPROM; Secondary Memory: Magnetic disk, Hard disk, Floppy disk, Optical disk, Compact disk (CD-ROM) and Solid state storage devices.

UNIT-III

Software: Introduction, Types of Software: Application software, System software. Operating system, functions of operating system, types of operating system. Data processing, Data processing systems:

batch processing, online processing, time sharing, real-time applications, Single-user, multi-user, and client-server systems; distributed and parallel processing systems; Translators: compilers, interpreters and assemblers.

UNIT-IV

Computer Networks: Introduction, types of networks on the basis of area coverage: LAN,

WAN, MAN. **Internet and WWW:** Evolution of Internet, Various Internet services (WWW, e-mail, telnet, ftp, IRC, news) and their uses, Access Methods, Browsers, Future of Internet, Applications of Internet, Evolution of www.

Suggested Resources for Reading:

1. 'Computers Today' by S.K. Basandra, Galgotia Publications.
2. 'Computer Fundamentals' by P.K.Sinha, N.D.: BPB. Publications.
3. 'Fundamentals of Computers' by V. Rajaraman, N.D.; PHI Publications.

References Resources:

1. 'Using the Internet' by Barbara Kasser, PHI, 4th ed., New Delhi.
2. 'Using the World Wide Web' by David A. Wall, PHI, New Delhi.

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Paper Title: COMPUTER HARDWARE Paper Code: HWN-103 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objectives: Hardware Fundamentals enable the students to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to computer hardware: Peripheral devices of a Computer system, Add On cards: network interface card, sound card and graphics card, functional description of various parts of a PC, UPS, Types of UPS: offline, online and line interactive UPS

UNIT-II

Various Components of a PC: Mother Board: Types of Motherboard AT, ATX, and BTX. Ports: Types of Ports, serial and parallel ports. Hard Disk: Types of Hard Disk: PATA, SATA, SCSI. RAM: Types of RAM: SRAM, DRAM, SDRAM, Power supply unit, cabinet, Processor.

UNIT-III

Cables: Types of Cables: USB, VGA, DVI, RJ11, RJ45, HDMI, SERIAL, and PARALLEL. Connecting Cables from SMPS to motherboard, hard disk etc. Establishing data connection for mother board, hard disk, and drivers.Fixing wires for power restart switches, fixing wires for power & HDD LED, External USB and Audio Connections. Drivers: Types of drivers: Device drivers, LAN drivers, sound drivers, graphics drivers.

UNIT-IV

BIOS: Introduction, Connecting & disconnecting computer peripherals and components Mouse, Keyboard, Monitor, Hard Disk. Window installation, Hard disk: partitioning and formatting, creating,

formatting and deleting partitions using DISKPART in Command Prompt and Graphical User Interface. Antivirus: features and Significance, Installing third party application. Firewall: Introduction to firewall, Features of firewall.

Practical based on HWN-103

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Govindarajalu, B., IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Tata McGraw Hill, 2nd ed. N.D.
2. Basandra, S. K., Computers Today, Galgotia Publications
3. Craig Zacker., The Complete Reference: PC Hardware, TMH Publication

References Resources:

1. Scott Mueller., Upgrading and Repairing PC, Pearson
2. Jean Andrew., A+ Guide to Hardware, 6th Edition

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Paper Title: NETWORK FUNDAMENTALS Paper Code: HWN-104 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objectives: Networking Fundamentals teaches the basic concepts and terminology of networking. The syllabus covers media types and standards. Students are also introduced to the terminology and basic concepts and Open Systems Interconnection (OSI) model.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Network concepts: Introduction to Network, characteristics, need, types of networks on the basis of area coverage: LAN, WAN, MAN; Advantages and Disadvantages of networks; Network Topologies: bus topology, ring topology, star topology, tree topology, mesh topology and hybrid topology. Network Models: Peer-to-Peer, protocol, client/server network and hybrid type network. Servers: Different types of servers, Network protocols

UNIT-II

Introduction to network standards: OSI reference model, TCP/IP Model. Difference between OSI reference model & TCP/IP model, Transmission Media: Types, Guided Media: twisted pair cable, coaxial cable, fibre optics, Unguided Media: radio wave, microwave, light wave, infrared and satellite.

UNIT-III

Networking protocols: DNS, FTP, Telnet, DHCP, HTTP, TFTP, SSH, SFTP, SMTP, POP, IMAP, NTPNetwork devices: HUB, Switch, Repeater, Bridges, Routers, Gateway: IEEE Standards: IEEE Standards for LAN and WAN: IEEE Standard 802.3 and Ethernet, IEEE Standard 802.4 Token Bus, IEEE Standard 802.5 Token Ring, IEEE Standard 802.6 Distributed Queue Dual Bus, IEEE Standard 802.2 Logical Link Control.

UNIT-IV

IP address: Introduction, IP address format, classes of IP Address, subnet address, Configuration of IP Address in PC, Configuration of IP address in Switch, configuration of Wireless Devices, Access Points Network Interface card (NIC): working principle of NIC, MAC Address, User Management: overview of user account, managing user account and password security: adding user, setting and changing password, changing picture of user, deleting the user, setting account restriction for the user, renaming a local account.

Practical based on HWN-104

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Tannenbaum, A.S., 2003: Computer Networks, Prentice Hall.
2. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education
3. Stallings, William, Local and Metropolitan Area Networks: Macmillan Publishing Co.

References Resources:

1. Black: Data Network, Prentice Hall of India.
2. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
3. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

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Paper Title: PRACTICAL ON NETWORKING-IPaper Code: HWN-105 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objectives: The objective of this paper is to provide in-depth knowledge about the concepts of networking. It offers integrated and comprehensive coverage of networking tools and devices.

Instructions for Examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT-I

Networking Tools: Crimping tool, punching tool: Impact and Punch down Tool, Multi Punch Tool, RJ45 Punch Down Tool, Impact Tool, Computer Punch Tool – deluxe, Cable cutter, I/O device, I/O Face Plate, I/O Box, LAN Cards, Connectors: RJ-11 (Registered Jack), RJ-45 (Registered Jack), F-Type, ST (Straight Tip) and SC (Subscriber Connector or Standard Connector), Fiber LC (Local Connector), MT-RJ (Mechanical Transfer Registered Jack), USB (Universal Serial Bus), LAN Patch Panel, Network Switch Rack

UNIT-II

Testing and Networking Tools: Introduction, Types of tools: LAN Testers, Application tools: IP scanner, Test Network Speed, Bandwidth Monitor 2 Lite Edition, Netbrute Scanner, piecework's IT Desktop, Ping, ipconfig, fstat, iproute, route, netstat, ipconfig /release, ipconfig /renew

UNIT-III

Working of cables: Coaxial cable, Unshielded twisted pair (UTP), Types of UTP Cable, Shielded Twisted-Pair Cable, Fiber optics, Patch Cords: UTP Patch cords, Fiber patch cords, Cable Manager, Cable Tag, Cable stripper, Scissors.

UNIT-IV

Working on Hub and Switch: Types of Hub: Active Hub, Passive Hub and Intelligent Hub, Types of switches: Unmanaged Network Switches, Managed Switches, Smart Switches.

Setting up of network: UTP cables, Switch, LAN tester, Switch Rack, Patch Panel, Cable Manager, Cable tagging, I/O box, I/O face plate, patch cords, connectors, terminals.

Suggested Resources for Reading:

1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
2. Cisco Networking Essentials, Author: mcmillan, troy|author; John Wiley & Sons
3. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
4. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education

References Resources:

5. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
6. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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B.Voc. (HARDWARE AND NETWORKING) Semester :II

Paper Title: SOFT SKILLS AND PERSONALITY DEVELOPMENT Paper Code: *GEN 201
Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objective: To expose the students to the concept of 'Human Development' (Personal and interpersonal) with emphasis on the latent resources that every human being possesses. To fulfil the need and importance of creating an awareness of these resources and to maximize the same to enable the students meet the challenges of the modern world

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Attitude, Process of attitude formation, How to build a success attitude, Spot analysis, Self management techniques, Self-image and self-esteem, Building self -confidence, Power of irresistible enthusiasm, etiquettes and manners in a group, public speaking, oral and written communication, Body language, Importance of listening and responding, tips for technical writing.

UNIT II

Development of Communication &Co-operation.Functions of Communication, Communication Basics, Communication Networks, Tips for Effective Internal Communication.Non-verbal Communication. Ethical Communication: austerity in speech, value, ethics and communication. Communication aids. Suitable behavior towards customers, Influence in skill, Creativity in presentation & projection and Multi-cultural skills

UNIT III

Introduction to Personality-Basic of Personality, Human growth and Behaviour, Theories in Personality, Motivation; Techniques in Personality development – Self-confidence, Mnemonics, Goal setting, Time Management and effective planning, Techniques in Personality Development-Stress Management, Meditation and concentration techniques, Self hypnotism, Self-acceptance and Self-growth.

UNIT IV

Co-ordination while working in a team, Leadership styles, Leader & Team player, Management of conflict, Profiles of great and successful personalities, Role of career planning in personality development, How to face personal interviews and group discussions

Suggested Resources for Reading:

1. 'Personality Development' by Rajiv K. Mishra, Rupa& Co
2. 'An Approach to Communication Skills' by Indrajit Bhattacharya, Delhi : DhanpatRai
3. 'Business Communication Skills' by Varinder Kumar, Bodh Raj, Manocha, Kalyani Publishers, New Delhi

References Resources:

1. 'Introduction to Psychology' by Atkinson and Hilgard's ,Edward E. Smith, Susan Nolen-Hoeksema, Barbara Fredrickson, Geoffrey Loftus
2. 'Communication Today & Tomorrow' by Ravi Aggarwal, Sublime Publications, Jaipur

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Paper Title: BUSINESS ETHICS **Paper Code: GC-202** **Credits: 6**

Job Role: Field Technician-Computing and Peripherals

Objectives: The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT 1

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics.

UNIT-II

Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics.Ethical Issues in Indian Business.

UNIT - III

Professional Values: Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality.

UNIT - IV

Corporate Social Responsibility: Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit, Corporate Social Responsibility and Corporate Governance.

Suggested Resources for Reading:

1. J.P. Sharma, Corporate Governance, Business Ethics & CSR, Ane Books Pvt. Ltd., Nee Delhi.

2. Andrew Crane, Dirk Matten, Business Ethics, Oxford University Press, New Delhi.
3. Daniel Albuquerque, Business Ethics, Principles and Practices (Indian Edition), Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, Business Ethics and Social Responsibility, Rexestore.

References Resources:

1. O. C. Ferrell, John Fraedrich, Linda Ferrell, Business Ethics: Ethical Decision Making & Cases, Cengagae Learning
2. Michael Blowfield, Alan Murray, Corporate Responsibility – A Critical Introduction, Oxford University Press, New Delhi.

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Paper Title: PERSONAL COMPUTING SOFTWARE Paper Code: HWN-203 Credits:6

Job Role: Field Technician-Computing and Peripherals

Objectives: The objective of this course is to familiarize students with Fundamentals of Information Technology and its applications. It enables the student to get practical exposure towards MS- Office tools

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT- I

DOS : Versions of DOS: Booting sequence; Warm and Cold reboot; Concept of File and directory , Redirecting command input and output pipes, Wildcard characters, Types of DOS commands: Internal and External; Internal Commands: DIR, MD, CD , CLS, COPY, DATE, DEL, PATH, PROMPT, REN,

RD, TIME, TYPE, VER, VOL; External Commands: XCOPY, ATTRIB, BACKUP, RESTORE, FIND, SYS , FORMAT, CHKDSK, DISKCOPY, LABEL, MOVE, TREE, DELTREE, DEFRAG, SCA NDISK, UNDELETE. Batch Files: Introduction to simple batch files; Introduction to CONFIG.SYS and AUTOEXEC.BAT files. Graphical User Interface: Fundamentals of windows, types of windows, anatomy of windows, Icons, Recycle bin Operations on window: Opening a Window, Minimizing and Maximizing a window, Moving window, Resizing Window, Closing the window windows explorer Folders: Creating and deleting folders, copying, renaming folders, folder properties. Control panel.

UNIT- II

Word Processing Package: Basics of Word Processing; Opening and Closing of documents; Text creation and Manipulation; Finding and replacing text, Printing of document, Formatting of text; Margin setting, Adding Borders and shading, Adding Headers and Footers, Setting up Multiple columns, Working with tables, Spell check, Grammar facility, Auto text, language setting and thesaurus; Mail merging. Installation of Word Processing Software.

UNIT- III

Spreadsheet Package : Worksheet Basics, Data Entry in Cells : Entry of numbers, text and formulae, Moving data in a worksheet, Moving around in a worksheet, Selecting Data Range, Using the Interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing; Formatting and Calculations: using Auto fill, Working with Formulae, Efficient Data Display with Data formatting (number formatting, date formatting etc.), Working with Ranges, Worksheet Printing; Working with Graphs and Charts : Creating Embedded Chart using char wizard, sizing and moving parts, updating charts, Changing chart types, Chart wizard, Adding Titles, Legends and Gridlines, Printing Charts; Database Management. Finding records with Data form, Adding/Deleting Records, Filtering Records in a worksheet; Functions and Macros: Worksheet Creating Macros, Recording Macros, Running Macros, Assigning Macros to Buttons, Defining Macros from Scratch. Multiple Worksheets.Installation of Spreadsheet software.

UNIT- IV

Presentation Packages: Basics, General Features, Creating a presentation, formatting and enhancing text, Incorporation of Animation, adding charts, multimedia, page setup and printing slides. Installation of Presentation software. Internet and WWW: Evolution of Internet, services provided on Internet, Access Methods, application of Internet.

Practical based on HWN-203

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Mathur Rajiv, 1996: Learning Word 6 for Windows Step-by-Step, Galgotia.
2. Mathur Rajiv, 1996: Learning Excel 5 for Windows Step-by-Step, Galgotia.
3. Jamsa, Kris A., 1993: Rescued by Windows 3.1, Galgotia. 5. Basandra, S.K., 1995

References Resources:

1. Computers Today, Galgotia.
2. Kasser, Barbara, 1998: Using the Internet, PHI, 4th ed., New Delhi.
3. Wall, David A. & Others, 1996: Using the World Wide Web, PHI, 2nd ed., New Delhi.

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Paper Title: COMPUTER HARDWARE AND TROUBLE SHOOTING

Paper Code HWN-204 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objectives: This paper enables the students to identify the hardware in-depth and helps them to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT- I

Trouble Shooting Basics: Basic Trouble shooting, the need for trouble shooting plan, elements of trouble shooting, general trouble shooting process. Diagnostic, Trouble Shooting and Repair Tools: Diagnostic software, Diagnostic hardware, the PC toolkit. Keyboard: Types of keyboard, correcting problems, vacuum cleaners and keyboards, keyboard maintenance, Mouse: Mouse basics, working with a mouse, performing mouse actions, inside the mouse, adjusting mouse properties, mouse keys, mouse maintenance, Optical mouse, breakdown and break through.

UNIT- II

Monitors: basic, VGA, DVI, viewable area, maximum resolution and dot pitch, power consumption, flat panel, monitor resolution, colour depth, degaussing, power option for monitor, display option for monitor, monitor maintenance. Hard Disk Drives: History, construction and operations of hard disk, read/write heads, File allocation table (FAT, FAT12, FAT16), Virtual FAT (VFAT), 32-Bit FAT (FAT32), New Technology File System (NTFS), High Performance File System (HPFS), Hard Disk tools: clean-up, error checking, disk fragmentation, change drive name and letters, virus scan, formatting a hard disk from windows, formatting a hard disk from DOS windows, installing master slave drives, CMOS identification of master slave disk, breakdown and break through.

UNIT- III

Processor: Basics, performance factors, modes, instruction set, CISC, RISC, and Superscalar. Processor associates: clock, motherboard, system bus, cache memory, instruction pipelining. Printers: printer characteristics, type quality, print speed, impact vs nonimpact, and fonts. Inkjet printers: common features, operations, parts of inkjet printer, control panel, trays, cartridges, cleaning cartridges, replacing cartridges, clearing paper jams. Laser Printers: common features, operations, technology, control panel, trays, output bin, cartridges, redistributing toner, internal cleaning, paper jams, cancel printing jobs, setting preferences and defaults of printing, breakdown and break through.

UNIT- IV

Mother board: Introduction, types, processor sockets and slots, memory sockets, cache and/or cache sockets, I/O bus slots, power connector, voltage regulator, capacitors, keyboard and mouse connectors, chipset chips, keyboard controller, real-time clock, super I/O controller, BIOS chips, battery, jumpers, ports and headers, pin connectors, upgrading motherboards, installing RAM and new processor. Power and SMPS: Power supply, AC-DC voltage conversion, parts of power supply, trouble shooting.

Practical based on HWN-204

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Govindarajalu, B., IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Tata McGraw Hill, 2nd ed. N.D.
2. Basandra, S. K., Computers Today, Galgotia Publications
3. Craig Zacker., The Complete Reference: PC Hardware, TMH Publication

References Resources:

1. Scott Mueller., Upgrading and Repairing PC, Pearson
2. Jean Andrew., A+ Guide to Hardware, 6th Edition

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Paper Title: Practical on Networking–II Paper Code: HWN-205 Credits: 6

Job Role: Field Technician-Computing and Peripherals

Objectives: The objective of this paper is to provide in-depth knowledge about the concepts of networking. It offers integrated and comprehensive coverage of networking tools and devices.

Instructions for Examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT- I

Patching of LAN Cable: Strip cable end, Untwist wire ends, Arrange wires, Trim wires to size, Attach connector, Check, Crimp, and Test. Setting up of LAN: UTP cables, Switch, LAN tester, Switch Rack, Patch Panel, Cable Manager, Cable tagging, I/O box, I/O face plate, patch cords, connectors, terminals.

UNIT- II

Internet connectivity: Different types of internet connection: Wireless, mobile, Hotspot, Dial-up, Broadband, DSL, Cable, Satellite, ISDN, Internet Service Provider (ISP), bandwidth or speed tester, Internet connectivity switch and PC, configuration of switch, managing of switching, troubleshooting of switching, testing application for switch.

UNIT- III

Wi-fi Connectivity: Basic Configuration of Wi-Fi device or access point, IP address configuration, DNS Address, DHCP mode, Security and Password, Admin login and password, MAC Filtering, Monitoring, Backup and restore, etc.

UNIT- IV

Switch connectivity: Managed and unmanaged switch, Basic configuration of switch, mac filtering, IP address configuration, DNS address, DHCP Mode, Admin login and Password, VLAN, backup file and restore settings.

Suggested Resources for Reading:

1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
2. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
3. Tannenbaum, A.S., 2003: Computer Networks, Prentice Hall.
4. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

References Resources:

1. Stallings, William, Local and Metropolitan Area Networks: Macmillan Publishing Co.
2. Black: Data Network, Prentice Hall of India.

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B.Voc. (HARDWARE AND NETWORKING) Semester :III

Paper Title: VALUE EDUCATION AND HUMAN RIGHTSPaper Code: *GEN 301 Credits: 6

Job Role: Network Support Engineer

Objective: The objective of this paper is to impart basic human values to students through formal education and contribute to making the student a true human being, who is able to face life and make it meaningful.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT -I

Value Education: Its purpose and significance in the present world; Value system – The role of culture and civilization; Holistic living – Balancing the outer and inner –Body, Mind and Intellectual level- Duties and responsibilities.

UNIT- II

Salient values for life- Truth, commitment, honesty and integrity, forgiveness and love, empathy and ability to sacrifice, care, unity , and inclusiveness, Self-esteem and self-confidence, punctuality – Time, task and resource management – Problem solving and decision making skills- Interpersonal and Intra personal relationship – Team work – Positive and creative thinking

UNIT- III

Human Rights – Universal Declaration of Human Rights – Human Rights violations – National Integration – Peace and non-violence – Dr. A P J Kalam's ten points for enlightened citizenship – Social Values and Welfare of the citizen – The role of media in value building.

UNIT- IV

Social Evils – Corruption, Cybercrime, Terrorism – Alcoholism, Drug addiction – Dowry – Domestic violence – untouchability – female infanticide – atrocities against women- How to tackle them

Suggested Resources for Reading:

1. 'Values and Ethics for Organizations: Theory and Practice' by Chakravarthy, S.K. , Oxford University Press, NewDelhi , 1999.
2. 'Ethics, Education, Indian Unityand Culture' by Satchidananda, M.K., Ajantha Publications, Delhi, 1991
3. 'Social Values among Youngadults: A changing Scenario' by Das, M.S. & Gupta, V.K. , M.D. Publications, NewDelhi, 1995

References Resources:

1. 'Education and Human Values' by M.G.Chitakra, A.P.H. Publishing Corporation, New Delhi, 2003
2. 'Humanist Values: A Source Book' by Bandiste, D.D., B.R.Publishing Corporation, Delhi, 1999
3. 'Human Values and education' by Ruhela, S.P. , SterlingPublications, New Delhi, 1986
4. 'Values and Education in Independent Indian' by Kaul, G.N.,Associated Publishers, Mumbai, 1975
5. 'Education in Values', NCERT, New Delhi, 1992

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Paper Title: E-COMMERCE Paper Code: GC 302 Credits: 6

Job Role: Network Support Engineer

Objective: The objective of this paper is to provide fundamental knowledge to the students about E-Commerce so that they can better perform in any area of operation and can excel in the field of commerce with IT specialization.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Electronic Commerce Framework, Basics and Tools of E-Commerce, Comparison of Web-based with Traditional Business; Growth of E-Commerce – Present and potential. E-Business: Meaning, Importance, Models Based on the Relationships of Transacting Parties (B2B, B2C, C2C and C2B), Present Status of E-Commerce in India

UNIT II

Changing Structure of Organization – The Impact of E-Commerce on supply chain management and customer relation, Socio-Economic Impacts of E-Commerce. Electronic Payment System: Types of Payment System — E-Cash and Currency Servers, E-Cheques, Credit Cards, Smart Cards, Electronic Purses and Debit Cards , Risk and Electronic payment Systems.

UNIT-III

Electronic Data Interchange (EDI): EDI and its applications in business; Legal, Security and Privacy issues in EDI, EDI standardization; Internet based EDI, Internal Information Systems, Work-flow Automation and Coordination, Types of Digital Documents, Digital Signatures.

UNIT-IV

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP, ERP Modules and design alternatives, challenges of ERP implementation, Business Process Re-engineering, ERP system development process. Regulatory aspects of E-commerce.

Suggested Resources for Reading:

1. 'Electronic Commerce, A Manager's Guide to E-Business' by PragDiwan and Sushil Sharma, Vanity Books International, Delhi.
2. 'Frontiers of Eletronic Commerce' by Ravi Kalakota, and Andrew B.Shinston, Addision Wesley.
3. 'Web Commerce Technology Handbook' by Minoli and Minoli, Tata McGraw Hill, New Delhi.

References Resources:

1. 'Electronic Commerce' by GrayP.Schneider, Course Technology, Delhi.
2. 'E-Commerce-The Cutting Edge of Business' by Kamlesh Bajaj &Debjani Nag, Tata McGraw Hill 2000.
3. 'Understanding E-Commerce' by David Kosiur, Microsoft Press. 1997

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Paper Title: WINDOWS OPERATING SYSTEMPaper Code: HWN-303 Credits: 6

Job Role: Network Support Engineer

Objective: This course enables students to understand the concepts of Operating System. After the completion of this paper, student will be able to manage various processes, files and folders on the disk and desktop with effective outcome.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT- I

Operating System: Introduction, features, Windows Operating System: features, history, hardware requirements, And installation of Windows Operating System Graphics interface: Benefits, Screen attributes: icons and bars. Mouse vs. keyboard input Features and accessories of the Windows program. Objects and their properties. Program management: Creating, saving and editing documents, Concurrent execution of programs, sharing data between programs.

UNIT- II

Folder and file management: Working with files, Naming files, Navigate to Folders with Windows Explorer, Copying and moving files, Deleting files, Managing folders, Creating, Viewing, Expanding and collapsing, Backing up and restoring files. Components of Windows: format of a window, moving windows, resizing windows, minimizing and maximizing windows.

UNIT- III

Control panel: Customizing screens, Screen colours, Patterns, Spacing icons, selecting time/date, setting the Sound, Concept of menu Using Help, Creating Short cuts, Basics of Window Setup, Notepad, Window Accessories, System restore. Customizing printing, changing the print queue, configuring the

printer, Adding printers. Working with fonts: changing, removing, adding, customizing mouse and keyboard use.

UNIT- IV

System properties and the device manager Management tools, DOS sessions, Explorer, Memory configuration, Safe mode Install and uninstall applications, Setup/troubleshooting issues. Maintaining and optimizing disks: Disk Cleanup, Disk defragmenter, Compressing and uncompressing folders and files. Encrypting and decrypting folders and files.

Practical based on HWN-303

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Jamsa, Kris A., Rescued by Windows 3.1, Galgotia.
2. Basandra, S.K., Computers Today, Galgotia.
3. Mathur Rajiv, Learning Word 6 for Windows Step-by-Step, Galgotia.
4. Mathur Rajiv, Learning Excel 5 for Windows Step-by-Step, Galgotia.

References Resources:

1. Darril Gibson, Microsoft Windows Networking Essentials 1st Edition.

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Paper Title: ROUTING & SWITCHING **Paper Code: HWN-304** **Credits: 6**

Job Role: Network Support Engineer

Objective: This course provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, it enable the students to enhance career development skill in the field of networking.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT- I

Router: Introduction, components of routers, Types of Routers: Broadband Routers, Wireless Routers, Edge Router, Subscriber Edge Router, Inter-provider Border Router, Core Router, Wired and Wireless Routers. Functions of routers, router modes, Configuration of Router: names, passwords, password encryption, interfaces name, creating a login banner, saving configuration and erasing configuration.

UNIT-II

Routing: Introduction, Types of routing: static and dynamic routing, Static routing: Configuring a static route, permanent keyword, static routes and administrative distance, configuring default route. Dynamic routing: Dynamic Routing Protocols: RIPv1, RIPv2, IGRP, EIGRP and OSPF. Delivery Semantics: unicast, anycast, multicast, broadcast, geocast. Configuring and Verifying RIP, IGRP, OSPF, EIGRP.

UNIT-III

Switch: introduction, roles of switches in network, types of switches: managed and unmanaged switch. Working of switch in different layers. Configuration of Switch: Command Mode, Setting Host Names, Setting Passwords, Setting IP Addresses and Default Gateways, Setting Operational Speed, Switch Port Security, Sticky MAC Addresses.

UNIT- IV

Switching: Introduction, Switching Services, Bridging vs LAN Switching, Switch Functions at Layer 2, Definition of Spanning Tree Protocol (STP), Operations of STP, STP Port States, Definition of VLAN, Typing of VLAN: Static and Dynamic, VLAN Identification Methods, VLAN Trunking Protocol (VTP), VTP Modes of Operation, Routing between VLANs, Configuring Inter-VLAN Routing.

Practical based on HWN-304

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons

References Resources:

1. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
2. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
3. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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Paper Title: ROUTING CONFIGURATION & NETWORK CONNECTIVITY

Paper Code: HWN-305 Credits: 6

Job Role: Network Support Engineer

Objective: Routing technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instructions for Examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT- I

Functionality of Router and Switch: Describe basic routing concepts: packet forwarding, router lookup process, Process Switching/Fast Switching/CEF, Configure and verify utilizing the CLI to set basic Router configuration: hostname, local user and password, enable secret password, console and VTY logins, exec-timeout, service password encryption, Interface IP Address, loopback, banner, motd, copy run start.

UNIT-II

Configure and verify operation status of a device interface: Serial, Ethernet, Verify router configuration and network connectivity using, ping, extended, traceroute, telnet, SSH, shcdpneighbors. Describe the boot process of Cisco IOS routers, Configure and verify utilizing the CLI to set basic Router configuration, Configure and verify operation status of a device interface, both serial and Ethernet, Verify router configuration and network connectivity. IP Routing Technologies: Configure and verify routing configuration for a static or default route given specific routing requirements.

UNIT-III

Differentiate methods of routing and routing protocols, Static versus Dynamic, 4.6.b Link state versus Distance Vector, next hop, ip routing table, Passive Interfaces (how they work), Admin Distance, and split horizon, metric. Configure and Verify OSPF: Benefit of single area, Configure OSPv2, Configure OSPv3, Router ID, Passive Interface, Discuss multi-area OSPF, Understand LSA types and purpose.

UNIT-IV

Configure and verify inter VLAN routing (Router on a stick), sub interfaces, upstream routing, encapsulation, Configure SVI interfaces, Manage Cisco IOS Files, Boot Preferences, Cisco IOS Images (15), Licensing, Show license, Change license, Configure and verify EIGRP (single AS), Feasible Distance/Feasible Successors/Administrative distance, Feasibility condition, Metric composition, Router ID, auto summary, Path Selection, Load Balancing, Unequal, Equal.

Suggested Resources for Reading:

1. Lacoste, CCNP Routing and Switching TSHOOT 300-135: Official Cert Guide, Pearson India
2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education

References Resources:

1. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
2. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
3. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

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B.Voc. (HARDWARE AND NETWORKING) Semester : IV

Paper Title: ENVIRONMENTAL STUDIESPaper Code: * GEN 401 Credits: 6

Job Role: Network Support Engineer

Objective: The objective of the paper is to understand the basic concepts of the natural environment, to identify the different sources of pollution and their related effects, and to have awareness about the conservation strategies and laws followed in India.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Basics of environment: Introduction, concept of biosphere—lithosphere, hydrosphere, atmosphere. Ecology, types of ecosystems, natural resources-types, importance, conservation.

UNIT-II

Renewable and non-renewable resources.Concept of sustainable development. Biodiversity: Definition, hotspots, national parks, biosphere reserves, wildlife sanctuaries.

UNIT-III

Pollution: Air, water, soil, noise and marine, Natural and anthropogenic sources of pollution. Effects of different types of pollutions Primary and secondary pollutants.Radioactive and thermal pollution. Acid rain

UNIT-IV

Global environment issues: Ozone depletion, global warming and climatic change. Environmental conservation strategies and projects in India, Ganga action plan, Yamuna action plan, Project tiger etc. Environmental laws in India.

Suggested Resources for Reading:

1. 'Ecology - Principles and Applications' by Chapman and Reiss, Cambridge University Press, Cambridge, U.K., 2004
2. 'Basic Environmental Biotechnology' by G.E. Omenn, Plenum Press, N.Y., 1987.
3. 'Environmental Challenges and Solutions' by Robert Jr. Cabin.

References Resources:

1. 'Global Environmental Change and Land Use' by H.J.T.M Harmaan, Jan Verhagen, C.A. Rovers, Springer, 2003.
2. 'A Textbook of Environment Science' by Thakur Vidya, Scientific Publishers, 2012. •'Living in the Environment: Principles, Connections, and Solutions' by Jr., G. Tyler Miller, Brooks Cole.

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Paper Title: PROJECT MANAGEMENT Paper Code: GC-402 Credits: 6

Job Role: Network Support Engineer

Objectives: The objective of this paper is to provide knowledge to students about the essentials of undertaking projects in an organizational environment.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Concept of Project: Meaning, Characteristics, Classification of Projects, Project Life Cycle and its Phases. Project Manager: Roles and Responsibilities, Project Management as a Profession.

UNIT-II

Generating and Screening Ideas – Steps, Monitoring the Environment, Scouting for Project Ideas, Preliminarily Screening, Project Rating Index. Feasibility Studies – Technical, Financial, Economic, Social, Legal and Managerial.

UNIT-III

Project Appraisal Techniques: Objectives, Types and Methods. Project Risks: Meaning, Types, Measurement of Risk,.Decision Tree Analysis (Basic Concepts only).Project Evaluation – Meaning, Evaluation v/s Appraisal, Objectives, Types of Evaluation, and Techniques.

UNIT-IV

Project Organization and Control – Project Network Analysis (Basic concepts of PERT, CPM, Cost and Time over Run). Project Reporting: Meaning, Purpose, Process, Requirements of a Good Report, Methods, Principles of Good Reporting System.

Suggested Resources for Reading:

1. Project Management - Choudhary - TataMcGraw Hill Pub.
2. Project Management: The Managerial Process (Special Indian Edit.) -Clifford F Gray, Oregon State University.

References Resources:

1. Projects: Planning, Analysis, Selection, Financing, implementation and Review -Chandra, Prasanna.

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Paper Title: WINDOWS SERVER ADMINISTRATION Paper Code: HWN-403 Credits: 6

Job Role: Network Support Engineer

Objective: The objective of this paper is to provide practical knowledge to the students about the Network Server. This course is meant to examine Windows Server while giving maximum learning potential to new Windows Administrators and Windows Developers.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Network Operating Systems: Windows Server Editions, Linux, Unix. Terminal services: Dumb terminals, terminal server configuration, windows server as terminal server, advantages of terminal services. Windows Server: different ways to install network operating system, pre-installation requirements. Maintaining AD DS, Managing User and Service Accounts, Implementing a Group Policy Infrastructure, Managing User Desktops with Group Policy, Installing, Configuring, and Troubleshooting the Network Policy Server Role.

UNIT-II

Installing and Configuring a Network Policy Server: Implementing Network Access Protection, Implementing Remote Access, Implementing Direct Access by Using the Getting Started Wizard Lab : Deploying an Advanced Direct Access Solution Lab : Implementing VPN Lab : Implementing Web Application Proxy, Optimizing File Services, Configuring Quotas and File Screening Using File Server Resource Manager Lab: Implementing Distributed File System, Configuring Encryption and Advanced Auditing, Deploying and Maintaining ServerImages, Using Windows Deployment Services to Deploy Windows Server, Implementing Update Management, Monitoring Windows Server

UNIT-III

Microsoft Windows Server: Deploying and Managing Windows Server, Introduction to Active Directory Domain Services, Installing Domain Controllers, Managing Active Directory Domain Services Objects, Automating Active Directory Domain Services Administration, Implementing Dynamic Host Configuration Protocol, Implementing DNS, Implementing Local Storage, Implementing File and Print Services.

UNIT-IV

Windows Security: Firewalls, Spyware, viruses, User accounts security, Search strategies System performance, Troubleshooting tools, Windows registry. Administering Windows Server: Configuring and Troubleshooting Domain Name System, Configuring and Troubleshooting DNS, Maintaining Active Directory Domain Services.

Practical based on HWN-403

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Windows Server Administration Fundamentals by Microsoft Official Academic Course
2. Darril Gibson, Microsoft Windows Networking Essentials 1st Edition
3. Tom Carpenter, Microsoft Windows Server Administration Essentials

References Resources:

1. Jamsa, Kris A., 1993: Rescued by Windows 3.1
2. Galgotia. 5. Basandra, S.K., 1995 Computers Today, Galgotia.

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Paper Title: NETWORK ADMINISTRATION & TROUBLESHOOTING

Paper Code: HWN-404 Credits: 6

Job Role: Network Support Engineer

Objective: The goal of this course is to train students to an exceptionally high standards in network planning, implementation, administration and support.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Backing UP and Restoring IOS Software and Configuration: Boot system commands, IOS file system, backing up configuration to a TFTP server, restoring configuration to a TFTP server, backing up the IOS software to a TFTP server, restoring/upgrading the IOS software from a TFTP server, restoring IOS software from ROM Monitor mode using Xmodem.

UNIT-II

Password-Recovery Procedures and configuration Register:The configuration register: A visual representation, bit means, the boot field, console terminal baud rate setting, console line speed: CLI & ROM Monitor mode, password recovery procedures for router, password recovery procedures for switches.

UNIT-III

Basic troubleshooting: viewing routing table, determining the gateway of last resort, last routing update, OSI layer 3 testing, OSI layer 7 testing, interpreting the show interface command, using CDP to troubleshooting, show controllers command debug commands, time stamps, Operating system IP verification commands, the ip http server command, netstat command.

UNIT-IV

Security Device Manager: SDM connecting with CLI and GUI, SDM Express Wizard with no CLI pre-configuration, resetting the router to factory defaults, SDM user interfaces, configuration routing using SDM, SDM monitor mode, SDM to configuration a router as a DHCP server & client, SDM configuration NAT/PAT.

Practical based on HWN-404

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
2. Cisco Networking Essentials, Author: mcmillan, troy|author; John Wiley & Sons
3. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
4. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education

References Resources:

1. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
2. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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Paper Title: SWITCHING CONFIGURATION & NETWORK CONNECTIVITY

Paper Code: HWN-405 **Credits: 6**

Job Role: Network Support Engineer

Objective: Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instructions for examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT-I

IP Services: Configure and verify DHCP (IOS Router), Configuring router interfaces to use DHCP, DHCP options (basic overview and functionality), Excluded addresses, Lease time, Describe the types, features, and applications of ACLs, standard (editing and sequence numbers) extended, named, numbered, Log option. Configure and verify ACLs in a network environment, named, numbered, Log option, Identify the basic operation of NAT, purpose, pool, static, 1 to 1, overloading, source addressing, one way NAT, Configure and verify NAT for given network requirements.

UNIT-II

Configure and verify NTP as a client. Recognize High availability (FHRP), VRRP, HSRP, GLBP, and Configure and verify syslog, Utilize Syslog Output, Describe SNMP v2 and v3. LAN Switching Technologies: Determine the technology and media access control method for Ethernet networks, Identify basic switching concepts and the operation of Cisco switches: Collision Domains, Broadcast Domains, Ways to switch, Store, Forward, Cut through, CAM Table.

UNIT-III

Configure and verify initial switch configuration including remote access management: Hostname, management IP address, IP default-gateway, local user and password, enable secret password Console and VTY logins, exec-timeout service password encryption, copy run start, Verify network status and switch operation using basic utilities such as: ping, telnet, SSH. Describe how VLANs create logically separate networks and the need for routing between them, Explain network segmentation and basic traffic management concepts, Configure and verify VLANs.

UNIT-IV

Configure and verify trunking on Cisco switches: dtp (topic), auto-negotiation, Identify enhanced switching technologies: RSTP, PVSTP, Ether channels, Configure and verify PVSTP operation: Describe root bridge election, Spanning tree mode. Configure and verify Switch Port Security: Sticky mac, MAC address limitation, static/dynamic, violation modes, err disable, shutdown, protect restrict, shutdown unused ports, err disable recovery, assign unused ports in unused VLANs, putting Native VLAN to other than VLAN 1.

Suggested Resources for Reading:

1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons

References Resources:

1. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
2. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

B.Voc. (HARDWARE AND NETWORKING) Semester : V

Paper Title: - Critical Thinking and Elementary Statistics Paper Code: * GEN 501 Credits: 6

Job Role: Network Administrator

Objective: The objective of the paper is to provide knowledge to student about basics of statistics and to inculcate the habit of critical thinking.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction: Statistics as a Subject ; Statistical Data: Meaning and Types , Collection and Rounding of Data, Classification and Presentation of Data.

UNIT-II

Analysis of Univariate Data: Construction of a Frequency Distribution; Concept of Central Tendency, mean, median and mode.

UNIT-III

Dispersion and Their Measures; Time Series: Meaning, Components, Models, Fitting Linear and Quadratic Trend

UNIT-IV

Index Number: Meaning, Types, and Uses: Methods of Constructing Price and Quantity Indices (Simple and Aggregate).

Suggested Resources for Reading:

- Chou-YaLun: Statistical Analysis, Holt, Rinchart and Winslon.
- Levin, Richard I. & David S Rubin: Statistics for Management, Prentice Hall of India, Delhi.

References Resources:

1. Aczel, Amir D.: Complete Business Statistics, Irwin.

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Paper Title: INTRODUCTION TO RESEARCH METHODOLOGY AND REPORT WRITING

Paper Code: GC 502 **Credits: 6**

Job Role: Network Administrator

Objective: The objective of the module is to familiarize the students with basics of research methodology like types of research, data collection, sampling techniques and report writing.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Research - Meaning, Characteristics, Types, Process and Utility,. Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research.Importance of review of literature.The research proposal.

UNIT II

Primary and Secondary data.Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT III

Sampling Design and Sampling procedure. Sampling and non-sampling errors.Data analysis and interpretation.Role of computers in research.

UNIT IV

Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

Suggested Resources for Reading:

1. 'Research Methods and Techniques' by C.R. Kothari, Eastern Wiley Publications.
2. 'Business Research Methods' by A. Bryman, and E. Bell, Oxford University Press, New York.

References Resources:

1. 'Research Methodology A Step by Step Guide for Beginners' by R. Kumar, Pearson Education.

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Paper Title: LINUX OPERATING SYSTEM Paper Code: HWN-503 Credits: 6

Job Role: Network Administrator

Objectives: This course enables students to get familiar with Linux system, its commands, files & directories, shell programming, and system administration. Student will learn to Work in the Linux environment and administrate the Linux machine.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT – I

Introduction: Linux Operating System, features, Basic Architecture, features of Kernel and Shell. Linux File system - Boot block, super block, Inode table, data blocks, How Linux kernel access files, Linux standard file system. Structure of file system, Essential Linux commands - Commands for files and directories : cat, cd, ls, cp, md, rm, mkdir, rmdir, pwd, file, more, less, file comparisons – cmp&comm, View files, disk related commands, checking disk free spaces, chmod with its options, cal,date,who,tty, lp,stty.

UNIT –II

Filters and pipes: head, tail, wc, pr, cut, paste, sort, uniqe, grep, egrep, fgrep, tee. The process : shell process, parent and children ,process status, system process, multiple jobs in background and foreground, changing process priority with nice, premature termination of process, Mathematical commands- bc, expr, factor, units

UNIT –III

Creating and editing files with VI editor with their command options, Operators, text deletion, text movement, changing text, yanking text, filtering text, the ex-mode, moving text from one file to another.

Communication: The bulletin board –news, write, mesg, talk, mail, elm, pine, finger, vacation and connecting to remote machine.

UNIT -IV

System administration Common administrative tasks, identifying administrative files – configuration and log files, Role of system administrator, Managing user accounts-adding &deleting users, changing permissions and ownerships, Installation of Linux system– Linux Installation requirement, complete Procedure steps, Partitioning the Hard drive, System startup and shut-down process, init and run levels. File system mounting, lpstat, backup strategy, installing software on Linux.

Practical based on HWN-503

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Tim Parker: Linux Unleashed, Techmedia Publishing House.
2. Norton, P.: Complete guide to LINUX, Techmedia.
3. Komarinski, M.: LINUX System Administration Handbook, AW.
4. John Goerzen: Linux Programming Bible, IDG Books, New Delhi.

References Resources:

1. Venkateshmurthy, M.G.: Introduction to UNIX & Shell Programming, Pearson Education.
2. Tery Dawson, Gregor N. Purdy, Linux Network Administration Guide, Oreilly.

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Paper Title: VMware WorkstationPaper Code: HWN-504 Credits: 6

Job Role: Network Administrator

Objective: The objective of the paper is to provide students with the knowledge of virtual machines and how to create the virtual machine the concepts to manage them.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Installing Workstation: Installing Workstation Basics, Installing Debuggers, Upgrading Workstation, Workstation Window Tour, Setting Workstation Preferences, Getting Help, Finding Your License Key, Creating Virtual Machines, Preparing to Create a VM, Understanding Memory Options, Understanding Page Faults, Understanding Virtual Disk, Understanding Virtual Disk Options, Creating a VM, Working with Easy Install, Installing a Guest OS, Installing a Guest OS , Virtualizing a Physical Machine, Importing Virtual Machines, Importing Virtual Box VMs, Installing VMware Tools, VM Files,VMDK File Names

UNIT-II

Using Virtual Machines: Starting a Virtual Machine, Stopping a Virtual Machine, Stopping a Virtual Machine, Resuming a Virtual Machine, Closing VMs and Exiting Workstation, Drag / Drop / Cut / Paste, Shared Folders , Mapping a Virtual Disk, Adding a Printer, Using Removable Devices, Removable Devices and Lost Data, VM Snapshot Overview, Creating Snapshots , Snapshot Best Practices, Installing New Software, Recording a VM, Deleting a VM, Goodbye Teams Hello Folders, Converting a Team

UNIT-III

Manage and Configure VMs: Workstation Display Preferences, Unity Mode, Encrypting VMs, Moving VMs, Understanding the UUID, Understanding Cloning, Cloning VMs, Changing Hardware Compatibility, Exporting VMs. Configuring Devices:Configuring DVDs, Configuring USB Controllers, Managing the VHD, Compacting the VHD, Expanding the VHD, Defragmenting the VHD, The Virtual Disk Manager, Configuring Keyboard Features

UNIT-IV

VM Networking: Understanding VM Networking, Common Network Configurations, Changing Default Network Config, Understanding Bridged Networking, Understanding NAT, Understanding Host-Only Networking, Add a Virtual Network Adapter, Modify a Virtual Network Adapter, Disconnect a Virtual Network Adapter, Configure Bandwidth and Packet Loss, VM MAC Addresses, Changing MAC Addresses, Sharing and Remoting: Understanding Workstation Server, Shared Virtual Machines, Connecting to a Remote Server

Practical based on HWN-504

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University

Suggested Resources for Reading:

1. The Book of VMware: The Complete Guide to VMware Workstation, Brian Ward, No Starch Press, 2002
2. The VMware Workstation 5 Handbook, Charles River Media Networking/Security Series Networking & Security, Steven S. Warren, Charles River Media, 2005
3. Virtualization: A Beginner's Guide, DannielleRuest, Nelson Ruest, Tata McGraw-Hill

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Paper Title: ADVANCED ROUTING AND SWICHINGPaper Code: HWN-505Credits:6

Job Role: Network Administrator

Objective: Routing and Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instruction for examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT- I

IP Addressing (IPv4/IPv6): Describe the operation and necessity of using private and public IP addresses for IPv4 addressing, Identify the appropriate IPv6 addressing scheme to satisfy addressing requirements in a LAN/WAN environment.

UNIT- II

Identify the appropriate IPv4 addressing scheme using VLSM and summarization to satisfy addressing requirements in a LAN/WAN environment. Describe the technological requirements for running IPv6 in conjunction with IPv4, dual stack, Describe IPv6 addresses, global unicast, Multicast, link local, unique local, eui 64, auto-configuration.

UNIT- III

Network Device Security: Configure and verify network device security features: Device password security, enable secret versus enable, Transport, disable telnet, SSH, VTYs, physical security, service password.

UNIT- IV

Describe external authentication methods. Configure and verify a basic WAN serial connection, Configure and verify a PPP connection between Cisco routers, Configure and verify Frame Relay on Cisco routers, Implement and troubleshoot PPPoE.

Suggested Resources for Reading:

1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education

References Resources:

1. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
2. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
3. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media.

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B.Voc. (HARDWARE AND NETWORKING) Semester : VI

Paper Title: - ENTREPRENEURSHIP DEVELOPMENT PROGRAMME

Paper Code: *GEN 601Credits: 6

Job Role: Network Administrator

Objective: EDPs aim at training various target groups in entrepreneurial traits so that they obtain adequate information, motivation and guidance in setting up their own enterprises.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Entrepreneur: Definition, nature, and characteristics of entrepreneurship: Emergence of entrepreneurial class including women entrepreneurs: Theories of entrepreneurship: Socioeconomic environment and the entrepreneur. Characteristics of Entrepreneur Leadership, Risk taking, Decision making and business planning. Innovation and Entrepreneurship, entrepreneurial behaviour and motivation.

UNIT-II

Small Business as a Seed Bed of Entrepreneurship: Concept of business venture. The start-up process, Concept, Plan, Implementation, Initial Strategic Planning, Product and Marketing Scope, Legal and Tax consideration, Risk analysis and financial considerations. Profit Planning in Small Enterprise: Growth Strategies and diversification.

UNIT-III

Finance Management in current operations and expansion of capital. Role of Small Business in the national economy, National Policies for small business development, Governmental and Non-Governmental assistance.Contribution of Commercial Banks in promoting and servicing small business.Small business and modern technology.

UNIT-IV

Rules & Regulations: Governmental Policies and formalities in setting up a unit. Basic requirements regarding registration, Excise, Sales Tax, Factory Act, SSI exemptions: Licensing and Registration procedure; Appreciation of important provisions of Factory Act,

Shops & Commercial Establishment Act; Sales of Goods Act, Partnership Act; Contract Act; Income Tax, Sales Tax and Excise rules; Insurance.

Suggested Resources for Reading:

1. 'Project Engineering & Management' by A.K. Sinha, Vikas Publishing House Pvt. Ltd., 1983.
2. 'Project Planning, Financing, Implementation & Evaluation' by U. K. Srivastava, Indian Institute of Management, Ahmedabad, 1981.
3. 'Financial Management - An Analytical and Conceptual Approach' by Kuchhal, S. C., Chaitanya Pub. House, 1982.
4. 'Principles of Management Accounting' by Mohan &Goyal, Agra SahityaBhavan, 1982.
5. 'Management of Small Scale Industries' by Saroja, Seth Publishers, Bombay, 1979.

References Resources:

1. 'How to Succeed in Small Industry' by Vepa Ram K., Vikas Publishing House, New Delhi, 1984.
2. Bare Acts: Central Sales Tax Act, State Sales Tax Act, Central Excise Act and Customs Act.
3. 'Law and Practice of Income Tax in India' by Bhagwati Prasad, NavmanPrakashan, 1972.
4. 'A Text Book of Commercial Law' by S. S. Gulshan, S. Chand & Co., 1979
5. 'Industrial Relations' by B.P.Gupta, PHD Chamber of Commerce &Inds, 1986

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Paper Title: TOTAL QUALITY MANAGEMENT Paper Code: GC 602 Credits:6

Job Role: Network Administrator

Objective: The objective of the paper is to provide knowledge to students of the concepts of total quality management and to inculcate among them a concern for quality and customer satisfaction.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis, Basic concepts of Total Quality Management, Principles of TQM.

UNIT II

Quality Council and Statements, Barriers to TQM Implementation. Quality: Customer Perception and Satisfaction, Customer Complaints, Service Quality, Customer Retention, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT III

Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process, Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM)

UNIT IV

Latest Challenges of Quality, Six Sigma Concepts: application. Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System– Elements, Implementation of Quality System, Documentation, Quality Auditing; ISO 14000, Requirements and Benefits.

Suggested Resources for Reading:

1. 'Total Quality Management' by A.V. Feigenbaum, McGraw-Hill Publication.
2. 'Total Quality Management' by J.S. Oakland, Butterworth – Hcinemann Ltd., Oxford Press.
3. 'Quality Management – Concepts and Tasks' by V. Narayana&N.S.Sreenivasan, New Age International.

References Resources:

1. Total Quality Management for Engineers' by Zeiri, Wood Head Publishers.
2. 'Total Quality Management' by James R. Evans, Cengage Learning

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Paper Title: WIRELESS COMMUNICATION AND NETWORKING

Paper Code: HWN- 603 Credits: 6

Job Role: Network Administrator

Objective: This course will enable the student to study about wireless networking and communication along with different technologies.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Wireless communication systems: Introduction-Evaluation of mobile radio communications, wireless communication systems example, paging systems, cordless telephone systems, various wireless systems compression. Mobile wireless communication systems: 2nd generation cellular networks, 3rd generation wireless networks, wireless in local loop, wireless LAN, Bluetooth and personal area networks (PAN).

UNIT II

Cellular system design fundamentals: Allocation of spectrum, basic cellular system, frequency reuse, channel assignment strategies, handoff strategies, interference and system capacity, trucking and grade off service, improving coverage and capacity, cell splitting. Multiple access technique for wireless communications: introduction to multiple accesses, FDMA, TDMA, spread spectrum multiple access, SDMA, packet radio, capacity of cellular systems.

UNIT III

Wireless Networking: Difference between wireless and fixed telephone networks, development of wireless networks, fixed network transmission hierarchy, traffic routing in wireless networks, wireless data services, common channel signalling. Wireless WAN: mechanism to support at mobile environment, communication in the infrastructure , iIS-95 CDMA forward channel, IS-95 CDMA risers channel, packet and frame formats in IS-95,IMT -20000, forward channel in W-CDMA and CDMA 2000, reverse

channels in W-CDMA and CDMA -2000 GPRS and higher data rates, short messaging service in GPRS mobile application protocols.

UNIT IV

Wireless land: Historical overviews of the land industry, evolution of the wan industry, wireless home networking IEEE 802.11 the PHY layer, Mac layer wireless ATM, Hyperlink, Hyper Lan-2

Orthogonal frequency division multiplexing: basic principles of orthogonality single versus multi channel systems, OFDM block diagramOFDM signal mathematical representation.

Practical based on HWN-603

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Wireless Communications, Principles, Practice – Theodore, S. Rappaport, PHI, 2nd Edn., 2002.
2. Wireless Communication and Networking – William Stallings, PHI, 2003.
3. Wireless Digital Communications – KamiloFeher, PHI, 1999.
4. Principles of Wireless Networks – KavehPahLaven and P. Krishna Murthy, Pearson Education, 2002.

References Resources:

1. Wireless Communications – Andreaws F. Molisch, Wiley India, 2006.
2. Introduction to Wireless and Mobile Systems – Dharma PrakashAgarwal, Qing-AnZeng, Thomson 2nd Edition, 2006.

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Paper Title: FIREWALL TECHNOLOGY Paper Code: HWN-604 Credits:6

Job Role: Network Administrator

Objectives: This paper enable the students to learn to configure, install and administrate a firewall and handle functionality of firewall.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Firewall: Introduction, Why do we need Firewall, How Firewall works, Network Layer Firewall, Application Layer Firewall, Windows Firewall, Types of firewall: Hardware and Software Firewalls, Firewall Filtering Techniques: Packet Filter, Application Gateway, Circuit-level Gateway, Proxy Server. Installation of Firewall: key functions during installation, special options during installation, installing UTM, Basic Configuration, Backup Restoration.

UNIT-II

Web Admin: Web Admin Menu Button bar, lists searching in lists, dialog boxes, buttons and icons, object lists. Dashboard: dashboard settings, flow monitor, Management: System settings, organizational, hostname, time and date, scan settings, reset configuration or passwords. Web admin settings: general, access control, user rights HTTPs certificate, user preferences, Licensing, up2date, backup/restore, user portal, notifications, hardware and software requirements, status system status configuration, shutdown and restart.

UNIT-III

Definitions & Users: Network definitions, MAC address definitions, service definitions, time period definitions, users and groups, global settings Interfaces and routing, Network Services: DNS: global, forwarders, request routing, static entries, DynDNS, DHCP: Servers, Relay, DHCPv6 Relay, Static Mappings, IPv4 table, Network Protection: Firewall, Rules, Country Blocking, Country Blocking Exceptions, ICMP, Advanced, NAT: Masquerading, NAT, Advanced Threat Protection: Global, Intrusion

Prevention: Global attack patterns, Anti-Dos/Flooding, Anti-Portscan, Exceptions, Advanced, Server Load Balancing: Balancing Rules

UNIT-IV

Web Protection: Web filtering: some key differences, common tasks, migration, Global, Https, Policies: Filter action wizard, categories, websites, downloads, antivirus, Web filter Profile: Filter profiles, filter actions, parent proxies, Filtering options: exceptions, websites, bypass users, potentially unwanted applications categories, Logging & Reporting : View Log Files, Today's Log Files, Archived Log Files, Search Log Files, Hardware Daily, Weekly, Monthly, Yearly.

Practical based on HWN-604

Instructions for practical:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Suggested Resources for Reading:

1. Hucaby, "CCNP Security FIREWALL 642-618 Official Cert Guide", Pearson India
2. Burns, "CCNP Security IPS 642-627 Official Cert Guide", Pearson India
3. Wilkins, "CCNP Security Secure 642-637 Official Cert Guide", Publisher: Pearson India

References Resources:

1. Dameon D Welch-Abernathy, "Essential Check Point Firewall-1 NG: An Installation, Configuration, and Troubleshooting Guide", Pearson Education
2. "Key Features of Framework for Monitoring Hardware Firewall Functionality Using Nids", LAP Lambert Academic Publishing.

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Paper Title: IMPLEMENTATION OF ADVANCE IP ROUTING Paper Code: HWN-605 Credits:6

Job Role: Network Administrator

Instruction for examiner:

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three question
3. Each question carries 10 marks.
4. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
5. Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Objective: IP ROUTINGtechnology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development

UNIT-I

Implement an EIGRP based solution, given a network design and a set of requirements: Determine network resources needed for implementing EIGRP on a network, Create an EIGRP implementation plan, Create an EIGRP verification plan, Configure EIGRP routing , Verify EIGRP solution was implemented properly using show and debug commands, Document results of EIGRP implementation and verification
Implement a multi-area OSPF Network, given a network design and a set of requirements: Determine network resources needed for implementing OSPF on a network, Create an OSPF implementation plan , Create an OSPF verification plan , Configure OSPF routing , Verify OSPF solution was implemented properly using show and debug commands, Document results of OSPF implementation and verification plan

UNIT-II

Implement an eBGP based solution, given a network design and a set of requirements: Determine network resources needed for implementing eBGP on a network, Create an eBGP implementation plan , Create an eBGP verification plan , Configure eBGP routing , Verify eBGP solution was implemented properly using show and debug commands, Document results of eBGP implementation and verification plan .
Implement an IPv6 based solution, given a network design and a set of requirements: Determine network resources needed for implementing IPv6 on a network, Create an IPv6 implementation plan, Create an

IPv6 verification plan, Configure IPv6 routing, Configure IPv6 interoperation with IPv4,Verify IPv6 solution was implemented properly using show and debug commands, Document results of IPv6 implementation and verification plan

UNIT-III

Implement an IPv4 or IPv6 based redistribution solution, given a network design and a set of requirements: Create a redistribution implementation plan based upon the results of the redistribution analysis, Create a redistribution verification plan, Configure a redistribution solution, Verify that a redistribution was implemented, Document results of a redistribution implementation and verification plan, Identify the differences between implementing an IPv4 and IPv6 redistribution solution

UNIT-IV

Implement Layer 3 Path Control Solution: Create a Layer 3 path control implementation plan based upon the results of the redistribution analysis, Create a Layer 3 path control verification plan, Configure Layer 3 path control , Verify that a Layer 3 path control was implemented Document results of a Layer 3 path control implementation and verification plan, Implement basic teleworker and branch services, Describe broadband technologies, Configure basic broadband connections, Describe basic VPN technologies, Configure GRE, Describe branch access technologies

Suggested Resources for Reading:

1. CCNP: Routing, CCIE No Kevin Wallace, Sean Odom, Todd Lammle
2. CCNP Routing and Switching V2.0 Official Cert Guide Library, Author Kevin Wallace, David Hucaby, Raymond Lacoste, Wendell Odom, Publisher CISCO System, 2014
3. CCNP: ROUNTING and SWITCHING QUICK Reference, Author: Denise Donohue, Brent Stewart, Publisher Cisco press
4. CCNP: ROUTE 642-902, Author: Wendell Odom, Publisher: Cisco press

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